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REVIEW

Reducing infectious complications after allogeneic stem cell transplant

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ABSTRACT

Introduction: Infections remain a significant problem, in patients undergoing an allogeneic hematopoietic stem-cell transplant (HSCT) and efforts have been made over the years, to reduce the incidence, morbidity and mortality of infectious complications.

Areas covered: This manuscript is focused on the epidemiology, risk factors and prevention of infections after allogeneic HSCT. A systematic literature review was performed using the PubMed database, between November 2019 and January 2020, with the following MeSH terms: stem-cell transplantation, infection, fungal, bacterial, viral, prophylaxis, vaccines, prevention. The authors reviewed all the publications, and following a common revision, a summary report was made and results were divided in three sections: bacterial, fungal and viral infections.

Expert opinion: Different infections occur in the early, intermediate and late post-transplant period, due to distinct risk factors. Improved diagnostic techniques, pre-emptive therapy and better prophylaxis of immunologic complications, have reduced the morbidity and mortality of infections. The role of the gut microbiota is under careful scrutiny and may further help us to identify high-risk patients.

ARTICLE HISTORY

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KEYWORDS

HSCT; bacteria; virus; fungal; prophylaxis

1. Aim of the study and methods

A comprehensive search of the electronic MEDLINE database was performed using the Pubmed search service. The search was realized between November 2019 and January 2020. The search terms included: 'epidemiology', 'allogeneic stem-cell transplant', 'antibiotic schedule', 'febrile neutropenia', 'bacterial infections', 'resistant bacteria', 'fungal infections', 'antifungal prophylaxis', 'vaccines', 'viral infections', 'antiviral prophylaxis', 'antiviral treatment', 'immunocompromised hosts', 'epidemiology'. All the abstracts were reviewed, and the relevant articles selected. In addition, a reference list from the selected articles was used to examine additional articles not included in the first search. Overall, 55 articles for bacterial infections, 41 articles for fungal infection and 97 articles for viral infections were selected.

1.1. Introduction

Allogeneic hematopoietic stem-cell transplantation (HSCT) is considered a standard procedure in the treatment of several disorders including hematological malignancies, bone marrow failure syndromes, congenital immunodeficiencies enzyme deficiencies and hemoglobinopathies. HSCT, however, comes with a significant risk of morbidity and mortality, mainly related to regimen toxicity, infections and graft versus host disease (GvHD) [1]. Several epidemiologic studies suggest that between 30% and 80% of patients may develop a bacterial, fungal or viral infection, following an allogeneic HSCT [1–5].

The risk of infections depends on several factors [1–7] which include:

- intensity of the conditioning regimen (myeloablative, reduced-intensity or non myeloablative)
- degree and duration of neutropenia
- type of donor based on HLA-matching and donor relation
- source of stem cells (bone marrow, peripheral blood or cord-blood)
- patient's clinical conditions, comorbidities and iron overload
- patients' diagnosis and phase of the disease (early or advanced)
- GvHD prophylaxis

The combination of these factors will determine the speed and quality of hematologic and immune recovery and therefore the risk and type of infections. Patients' survival is critically related to prevention, early identification and prompt treatment of infections.

Prevention has been shown to be a key factor in minimizing infectious complications in patients undergoing HSCT. Several studies have focused on the protective environment, where patients are admitted (e.g. hospital logistics, room ventilation, air filtration), others on the behavioral aspects to be adopted by health-care workers, visitors and patients. HSCT recipients should be admitted in single bedrooms, with bathroom and ventilation (at least 12 air exchanges per hour). The

Article highlights

- Centres offering allogenic HSCT should document the incidence and epidemiology of infections in order to adjust therapy accordingly.
- The risk for invasive fungal infections in allogenic HSCT is dynamic. Re-evaluation of new appearing risk factors during the transplantation process is mandatory, to adapt prophylaxis when needed.
- HSCT centres should train patients and relatives for preventing infections in the hospital, at home and in the community.
- Baseline serostatus for recipient and donor should be carefully assessed and managed before starting the conditioning regimen.
- Prevention of acute and chronic GvHD, leading to a rapid immune recovery, remains the key issue for a safe post-transplant outcome.

room or the Unit should be equipped with central or point-of-use high-efficiency particulate air filters (HEPA). The patients' rooms should be under positive pressure to avoid infiltration of airborne pathogens from outside. Frequent monitoring of filtration efficiency is required to determine how often filters need to be replaced [8–12].

Regarding the behavioral aspects of nurses and physicians, one essential point is hand hygiene: this applies also to physicians, nurses and visitors, with the aim of reducing the risk of cross infections [13]. The patient's room should be cleaned more than once a day, by trained cleaning staff, and should be equipped with waterproof and antistatic furniture [14,15]. CDC guidelines also focus on additional key aspects for patients undergoing HSCT, such as cleaning clothes, screening policies for visitors, control of protective equipment (gloves, gowns, masks, respirators and eyewear). Finally, special attention should be given to the patient's personal hygiene, in order to reduce infections by endogenous organisms [9].

2. Phases of opportunistic infections among HSCT recipients

Hematologic and immune recovery depends on several factors, including the stem-cell source, the degree of donor/recipient HLA matching, and the severity of acute and chronic GvHD. The timing and type of post-transplant infections reflects the predominant defect of host defenses. There are three phases of hematologic/immune recovery: a pre-engraftment phase, from day 0 to day +30 post-HSCT; a post-engraftment phase, from day +30 to day +100 post-HSCT; and a late phase from day +100 post-HSCT onwards [16].

During the first phase, risk factors are the duration of neutropenia and the degree of the mucocutaneous barrier disruption, including mucosal injury and vascular devices. The post-engraftment phase is characterized by varying degree of impaired cell-mediated immunity, depending on the quality of engraftment, and the severity of GvHD. In the late phase, humoral and cellular immunity may have recovered, but are greatly influenced by chronic GvHD and its treatment [16,17].

3. Bacterial infections

BSI are reported to occur in 25–45% of HSCT recipients [18,19]. In a recent study performed at the Hospital Clinic in

Barcelona, changes in BSI epidemiology were collected during a 25 years period (from 1993 to 2017). Catheter-related BSI accounted for 45.1% of cases and endogenous/unknown sources for an other 39.6% [20]. Several variations may occur between different centers and among patients, due to different transplantation platforms, with different stem cell sources, conditioning regimens and GvHD prophylaxis [21]. In the pre-engraftment period, both Gram-negative and Gram-positive BSI can occur, while in the late phase, encapsulated pathogens are more frequent (*Haemophilus influenzae* and *Streptococcus pneumoniae*) [16]. Commonly, oral mucositis facilitates infections due to viridans Streptococci and Coagulase-negative Staphylococci, whereas enteric mucositis promotes infections caused by Enterobacteriaceae, Enterococci and *Pseudomonas aeruginosa*. Central venous catheters represents an additional risk factor [21].

Based on the literature review and a survey in HSCT recipients, on behalf of the European Conference on Infections in Leukemia (ECIL), Enterobacteriaceae were isolated in 30% of BSI, Coagulase-negative Staphylococci in 24% and Enterococci in 8%; in the remaining cases, the pathogens isolated were viridans Streptococci (6%), *Staphylococcus aureus* (5%), *Pseudomonas aeruginosa* (5%), other Gram-positive (5%) and other Gram-negative (5%) [22]. A decreased incidence of coagulase-negative staphylococci, with an increase of *Enterococcus faecium* were reported. Extended spectrum beta-lactamase (ESBL) producer Enterobacteriaceae and MDR-*Pseudomonas aeruginosa* has been, also, noted [20,23,24].

3.1. Gram positive

The mortality of gram-positive BSI is reported to be low (less than 5%) [25]. Enterococci are the cause of 10–12% of gram-positive BSI in HSCT patients. Enterococci infections usually occur in the first post-transplant period and are usually nosocomial. *Enterococcus faecium* is the predominant species. The incidence of vancomycin-resistant Enterococci (VRE) in Western Europe is relatively low in comparison to the incidence in the USA [26]. VRE bacteremia in HSCT recipients has significant consequences especially in neutropenic subjects, during the pre-engraftment phase: indeed, despite being considered to be a relatively nonvirulent pathogen, it can cause septic shock [27,28]. In order to prevent VRE infection, clinicians should carefully consider the use of vancomycin treatment and antimicrobial agents with anti-anaerobic coverage. In addition, active surveillance has also been recommended, with rectal and stool cultures to identify colonized patients, although, the effectiveness of active surveillance is controversial [29].

Staphylococci represent more than 30% isolated during BSI [22]. One of the most common cause of BSI is Coagulase-Negative Staphylococcus (CNS) infection, defined as two or more consecutive positive blood cultures. Recipients with a central venous catheter are particularly susceptible to CNS infections [30].

Methicillin-susceptible *Staphylococcus aureus* (MSSA), or methicillin-resistant (MRSA) are major causes of infections after transplantation [31]. According to the ECIL survey, MRSA were reported in more than 50% of isolated CNS infections, while the resistance for *Staphylococcus aureus* was reported to be lower [22]. Patients with methicillin-resistant *Staphylococcus* infection must be placed under contact precautions, until the discontinuation of antibiotic treatment and until three consecutive weekly cultures are reported to be negative [9]. However, also vancomycin intermediate *Staphylococcus aureus* (VISA) and heterogeneous or hetero-resistant VISA (hVISA) have been reported, although more rarely, in HSCT recipients [32].

HSCT patients are also frequently at risks of contracting *Streptococcus viridans* infections, which may be complicated in 7%–39% of cases, with acute respiratory distress syndrome (ARDS) or septic shock. Dental treatment and oral hygiene, 3–4 weeks before transplantation, may reduce the risk of infection [15].

Infections due to *Streptococcus pneumoniae* occur in the late phase after HSCT, with a mortality of 5% to 35% [33,34]. Vaccination against *Streptococcus Pneumoniae* should be given in the post transplantation phase, although the effectiveness will be influenced by the degree of immunodeficiency, which is in turn influenced by chronic GvHD [35].

HSCT recipients exposed to broad-spectrum antibiotics (which may change the microbiota) and to chemotherapy (damaging the intestinal mucosa), are susceptible to *Clostridium difficile* infections (5–30%). In general, HSCT recipients are at high risk of developing this infection, especially if they are pre-colonized [36,37]. Routine stool-culture within 3 days of admission is recommended in patients with diarrhea. Isolation must be considered once the diagnosis is established. A strict adherence to both hand hygiene and contact precautions plays a crucial role in significantly reducing the incidence of this infection [38,39].

3.2. Gram negative

In order to minimize Gram-negative infections, one may consider the following factors: colonized or previously infected patients are a major reservoir of Gram-negative bacteria and dissemination may occur via intermediate vectors, including infected hands of healthcare workers (HCW), and visitors. The transmission of Gram-negative bacteria, depends on the ability of bacteria to survive for long time on dry and moist areas.

An Italian prospective multicenter study has demonstrated that antibacterial prophylaxis during neutropenia, using fluoroquinolones, does not reduce the rate of Gram-negative infections in HSCT [40]. In several centers there is increasing evidence of antimicrobial resistance among pathogens, isolated in allogeneic transplant wards. This problem is relevant for multidrug resistance (MDR) Gram-negative infections: a mortality of 40% has been reported for MDR *Pseudomonas aeruginosa* and 64% in MDR *Klebsiella pneumoniae* [19,31,41].

Transmission of *Escherichia coli* occurs by fecal-oral contact, via contaminated food and/or water. ESBL-producing *E. coli*

transmission does not occur exclusively in hospital settings, but also in households. Several studies have shown that antibiotic prophylaxis in HSCT patients, particularly with fluoroquinolone, increases the resistance in *E. coli* [42–44].

Enterobacter spp. and *Serratia* spp. cause nosocomial outbreaks, in which environmental contamination, of both dry and moist surfaces, is the key to dissemination [45].

In the last decade, *Klebsiella pneumoniae* (MDR KP) has become a leading cause of nosocomial infections. In most cases, transmission occurs by hand-contact between HCWs which carry the infection from colonized patients, or environmental-infected surroundings, to new patients. Several studies have also highlighted outbreaks caused by contaminated food or by contaminated sinks [46]. A recent multicenter study on MRD-KP, reported a mortality rate of over 50% in patients with hematologic malignancies, including HSCT recipients [47]. Susceptible treatments include ampicillin/sulbactam, colistin and tigecycline [48–51]. In a retrospective case-control trial, Kim et al. reported a mortality rate of 95% for BSI due to carbapenem and MDR *Acinetobacter Baumannii* in HSCT recipients, requiring admission in intensive care units [52].

Salmonella spp. has also been described as a nosocomial pathogen. Most probably, transmission consists in food-borne-mechanism, however cross-transmission through HCWs hands may also occur [53].

3.2.1. General practice for infection prevention and control

Prompt identification of potential colonized patients on admission, through screening cultures

Isolation of infected/colonized patients in single rooms, with isolation signs, to reduce the risk of transmission to other patients (dedicated bathroom and/or dedicated water closet)

hand hygiene

implementation of environmental cleaning policies

dedicated antimicrobial stewardship program

recording, documentation and analyses of in house infectious trends

3.3. Specific preemptive and preventive recommendation

The National Comprehensive Cancer Network recommends high-dose levofloxacin doses (500–750 mg) in order to prevent Gram-negative and some Gram-positive pathogens [54]. However, a prospective multicenter study has demonstrated that antibacterial prophylaxis during neutropenia, using fluoroquinolones, does not reduce the rate of Gram-negative in HSCTs, and this should be kept in mind in the era of increasing bacterial resistance [40].

Immunizations with vaccines are recommended to prevent infections due to *S. pneumoniae*, *Haemophilus influenzae type b*, *Bordetella pertussis*, *Clostridium tetani*, *Neisseria meningitidis* and *Corynebacterium diphtheria* [55].

Encapsulated bacteria cause infections during the late post-engraftment period and the most important prophylactic measure is vaccination. Meningococcal vaccination is

recommended starting at 6 months post-transplant, at least 2 doses, with either a Men C or tetravalent vaccine and a Men B vaccine; pneumococcal vaccination, for patients with GvHD, includes PCV 13 (Prevenar) starting 3 months after transplant, 3 doses, given at monthly intervals and a fourth dose can be administered 6 months after the third dose [55]. In patients without GvHD pneumococcal vaccination should be supplemented with one dose of PPSV23, 12 months after transplantation, but not sooner than 8 weeks after the last PCV. *Haemophilus influenzae B* vaccination is recommended 3 months after transplant, in 3 doses of conjugate vaccine given at monthly intervals. Three doses of a DTP-Hib vaccine, 6 months post-transplant, can be considered as a valid alternative [55].

4. Fungal infections

Invasive fungal disease (IFD) is an important cause of morbidity and mortality in hematological patients, particularly after a HSCT. The development of improved antifungal prophylaxis strategies with newer-generation triazoles has resulted in improved outcome. However, this has occurred together with an increasing use of alternative donor transplants and older patients. The incidence of IFD during the first year post-transplant remains between 8 and 12% [56–63].

4.1. Epidemiology

The epidemiology of IFD depends on host and environmental factors. Pathogens can be divided into three groups: *Candida* species, *Aspergillus* species and other molds (including Zygomycetes, *Fusarium* species, and *Scedosporium* species). Over time, and with the introduction of prophylaxis for patients at risk, we have seen a shift from *Candida* species to *Aspergillus* species, as the most common fungal pathogen in patients receiving HSCT. The estimated cumulative incidence of proven-probable Invasive Aspergillosis during the first post-transplant year is 7%, while Invasive Candidiasis is 1% [5,57,64]. Other molds represented less than 0.5%. The Italian Group for Marrow Transplantation (GITMO) in a recent survey, reported that most infections occurred during the early phase (day 0–40) with a cumulative incidence of proven/probable IFD (PP-IFD) of 5.1%. They attributed that high rate of early-IFDs to the high number of patients who had 1 or more risk factors, as explained below [56].

4.2. Risk factors

Mismatched related and unrelated donor transplant recipients, have a higher risk of IFD, compared to HLA-matched transplant, with an incidence of 8.1%, 7.7% and 5.8%, respectively, [59]. In a large observational study, including 1401 patients, undergoing allogeneic or autologous SCT the cumulative incidence of IFD at 6 months from transplantation was 7.7%. The cumulative incidence in recipients of HLA-matched related, mismatched related, and unrelated HSCT was 4.3%, 13.2%, and 12.8%, respectively; it was significantly higher in recipients of cord blood (CB) transplants [57].

In a 2014 GITMO study, including 1771 HSCT recipients, there was a 1.9 fold increase of IFD in mismatched related transplants, 2.86 in unrelated donor (UD) grafts and 5.45 in unrelated cord blood grafts, as compared to matched related transplants [56].

4.3. Pre-HSCT risk factors

Pre-transplant risk factors for developing an IFD are a combination of host's comorbidities, hematological disease status, source of the graft and transplant platform.

Individual risk factors include older age, diabetes, use of central venous catheter, iron overload, prior IFD [65], and a previous lung disease (i.e. chronic obstructive pulmonary disease, occupational exposure, lifestyle). Older age is a significant risk factor for IFD: in a multivariate analysis on aspergillosis in the 2004 SEIFEM study, patients over 36 years, had a 5.2 increased odds of IFD, compared to the younger cohort [5]. Iron overload (with ferritin levels >2000 ng/mL) is an independent risk factor for invasive aspergillosis in HSCT recipients: indeed high levels of free iron may impair cellular antimicrobial defense and favor fungal growth [58,66]. Another described risk factor is genetic predisposition, with specific polymorphisms, including toll-like receptor 4 gene [67], long pentraxin 3 (PTX3) [68], and dectin-1 [69], causing an altered inflammatory response and an increased risk of IFD in HSCT.

An additional risk factor is the underlying hematological disease, with an increased risk of early IFD for acute myeloid leukemia (AML) and a high risk of late IFD for lymphoma. The association of IFD and lymphoma may be explained by the advanced phase of the disease, in patients eligible for allogeneic SCT, and by the numerous lines of chemotherapy and long periods of neutropenia [60].

The GITMO study found an elevated incidence of early IFD following an allogeneic SCT [56]. This high incidence was associated with the following risk factors: an IFD in the 6 months prior to transplantation, active acute leukemia as the underlying disease, an unrelated donor or cord blood graft. The incidence of early IFD was 16%, 12%, 12% and 6.4%, respectively, for cord unrelated, family mismatched and identical sibling grafts [56]. Patients with a previous diagnosis of invasive aspergillosis were at high risk of reactivation early after HSCT; therefore, secondary prophylaxis is strongly recommended [70].

4.4. Post-HSCT risk factors

Post-HSCT risk of mold infections is determined by the degree and duration of neutropenia and lymphopenia, the type of GvHD prophylaxis and by post-transplant events, such as acute or chronic GVHD and viral infections. Prolonged and profound neutropenia (ANC <500/mm³ for >14 days), as well as lymphopenia (ALC <300/mm³) and monocytopenia (AMC <100/mm³) are independent risk factor for IFD [57,58,71]. Patients with severe, grade III and IV acute GvHD, have a high risk of IFD, as compared to patients with grade II

GvHD [72]. The GITMO study [56] showed that the risk for IFD in patients with grade II–IV aGvHD was low, if patients received a matched-related donor graft (MRD) (2.3%), but was high in patients receiving mismatched related donor (MMRD), UD, and CB grafts (10%). When acute GVHD was followed by chronic GVHD, the risk was very high in patients receiving alternative donor grafts (25%). Of interest, the rate of IFDs in patients with chronic GvHD not preceded by acute GvHD was relatively low (< 4%) in all types of transplantation.

Steroid treatment also increases the risk of invasive aspergillosis: both high dose steroids or prolonged treatment with moderate doses (0.25–1 mg/kg/d for 2 to 10 weeks) can increase the risk for IFD [73]. Corticosteroids affect the host immune response to *Aspergillus* species by preventing alveolar macrophage killing of phagocytosed *Aspergillus fumigatus* conidia and by reducing alveolar macrophage production of proinflammatory cytokines. Other immunosuppressive drugs like anti-thymocyte globulins, alemtuzumab and antiCD25 (basiliximab) also increase the risk of IFD [57,58].

Cytomegalovirus reactivation is a classical risk factor for IFD after allogeneic SCT, and its occurrence is an indirect marker of severe immunosuppression [58]. It is frequently associated with GvHD, and may result in neutropenia which may be worsened by ganciclovir treatment. The association of influenza and other respiratory virus, with the development of pulmonary invasive aspergillosis, has also been reported [4,58].

4.5. Prevention

Prevention plays an important role and great efforts have been made, both with chemoprophylaxis and environmental measures, in order to minimize the risk of IFD.

4.5.1. Environment

There are differences in barrier nursing policies, among transplant centers, and the efficacy of some of them remains controversial. One prospective study showed a significant trend for less pneumonia, in HEPA-filtered rooms after HSCT [74]. Construction activity has been reported to be an independent risk factor for invasive fungal infections so nearby constructions should be kept to a minimum [75]. It is also recommended to avoid plants and soil in the patient room, as they serve as fungal reservoir and vascular access should be properly cared to avoid *Candida* infections. Some other general measures as hand washing, use of masks, controlled water and diet are also recommended, although there is a lack of appropriate literature on these topics [11].

Centers performing HSCT, should analyze their own incidence and epidemiology of IFD, in order to adjust prophylaxis and therapy accordingly.

4.5.2. Chemoprophylaxis

In the early nineties, amphotericin B deoxycholate was used as prophylaxis against IFD, but effectiveness was limited, and nephrotoxicity and infusion-reactions were common. Fluconazole instead has a good tolerance/efficacy ratio: two

randomized trials, comparing fluconazole (400 mg/day for 75 days) and placebo in HSCT recipients, showed significantly improved protection against candidiasis and survival in the fluconazole group, compared with placebo [76,77].

However, fluconazole has no effect against mold infections. For this reason prophylactic itraconazole was tested in two randomized trials [78,79]. The studies compared fluconazole with oral or intravenous itraconazole over 180 days after HSCT. In both studies, itraconazole provided better protection against mold infections, and similar protection against candidiasis; there was no difference in overall or fungal-free survival, and more patients in the itraconazole arm developed hepatotoxicity or discontinued treatment, because of gastrointestinal intolerance.

The appearance of newer-generation triazoles provide options for prophylaxis of invasive aspergillosis. Posaconazole was compared to fluconazole and itraconazole in one study in acute myeloid leukemia and myelodysplastic syndrome (AML/MDS) patients, and reduced the incidence of invasive aspergillosis [80]. Ullman's double-blind trial in 2007 compared also posaconazole and fluconazole for prophylaxis against fungal infections among patients with GVHD; posaconazole was superior in preventing invasive aspergillosis and reducing deaths related to fungal infections [81]. Several studies since then have confirmed the superiority of prophylactic posaconazole, compared to fluconazole in patients at high risk for IFD [82,83].

Some years later fluconazole was tested against voriconazole, given for 100 days post-transplantation, or 180 days in high-risk patients. They saw a trend toward fewer IFDs and *Aspergillus* infections with voriconazole, but fungal-free survival and overall survival was similar [84]. The incidence of severe adverse events was also similar. A second open-label study comparing voriconazole with itraconazole proved better tolerance to voriconazole with less treatment-related adverse events, and no difference in IFD incidence or survival to day 180 [85].

Drug–drug interaction and metabolic variability of voriconazole, expose the patient to the risk of either subtherapeutic levels or toxicity, suggesting the need for therapeutic drug monitoring [TDM]. Oral solution posaconazole, the only clinically available formulation for many years, is also highly variable and negatively influenced by several factors like lower absorption in patients with mucositis or GvHD of the gastrointestinal tract. This also makes TDM mandatory. Delayed-release tablet of posaconazole is now available for clinical use: administration of 300 mg posaconazole tablets once daily, is well tolerated, with a safety profile similar than the oral suspension [86]. In addition plasma concentrations of posaconazole administered as tablets, are significantly higher as compared to posaconazole oral suspension. Emergent aspergillosis occurred in 8.7% patients in the suspension group and in none of the patients receiving the tablet formulation [87]. Thus, the tablet formulation of posaconazole is safe, effective and provides predictable absorption.

One study evaluated pharmacokinetics and safety of intravenous posaconazole in antifungal prophylaxis of

neutropenic patients with AML, MDS or in the context of HSCT [88]. In total, 237 received 300 mg posaconazole i.v. twice daily on day 1 and thereafter 300 mg iv once daily for up to 28 days. In almost all patients a concentrations between 500 and 2500 ng/mL was reached. The main adverse events reported were diarrhea, nausea and rash (8, 5 and 5%, respectively). Further randomized trials are needed, but intravenous posaconazole could be an option when oral treatment is not appropriate/possible. Of interest, some clinical studies observed a prolonged 'protective effect' of posaconazole (compared with fluconazole or itraconazole) given in the induction phase of AML, with lower incidence of IFD also during consolidation and transplant procedures [89].

Data for the echinocandins are limited to micafungin (50 mg iv q24h). A randomized study showed superiority of micafungin versus fluconazole for antifungal prophylaxis in neutropenic patients [90]; however, the study had significant deficiencies and an overrepresentation of a low-risk population.

One recent prospective observational study comparing micafungin (50 mg iv q24h) and fluconazole showed a high incidence of IFDs caused by resistant fungi in the micafungin group [91]. Another recent open-label study in patients with AML/MDS randomized micafungin 100 mg iv daily or posaconazole suspension 400 mg orally twice daily: the primary endpoint was prophylaxis failure including premature discontinuation due to infection, intolerance, adverse event, or death. Prophylaxis failure occurred in 34.5% and 52.7% of patients on micafungin and posaconazole; micafungin failures were largely due to antifungal treatment, posaconazole failures were mostly due to gastrointestinal intolerance or adverse effects. IFD incidence and survival were similar between study arms [92]. The authors propose micafungin as a good alternative for antifungal prophylaxis in patients with neutropenia.

Isavuconazole has been approved for the treatment of invasive aspergillosis and mucormycosis but there is still

a lack of data on its prophylactic use. A retrospective review of breakthrough invasive fungal infections (bIFDs) among adult hematologic malignancy patients and HSCT recipients, the largest study to date, showed a trend toward an increased incidence of bIFDs during isavuconazole prophylaxis compared to posaconazole and voriconazole. Further studies are required, to determine the role of isavuconazole for primary prophylaxis [93].

4.6. Guidelines recommendations for antifungal prophylaxis

Every patient undergoing HSTC is considered as 'medium risk' for IFD, during the pre-engraftment period. Patients at high risk for early IFD are as following: active acute leukemia at the time of transplantation, cord blood transplantation, previous IFD, cortico-refractory/cortico-dependant aGvHD, and mismatched related donor (MMRD) or unrelated donor (UD) with steroid therapy of ≥ 2 mg/kg/day for at least 1 week, CMV disease, recurrent CMV infection, prolonged neutropenia ($<500/\mu\text{L}$ for more than 3 weeks, iron overload), or grade II aGvHD. A mold-active primary antifungal prophylaxis should be used in high-risk patients [56].

The recommendations from ECIL, European Society of Clinical Microbiology and Infectious Diseases (ESCMID), Infectious Diseases Society of America (IDSA) are reported in Table 1 [94–96]. The ECIL recommend in the pre-engraftment phase, fluconazole (400 mg/day) for centers with a low incidence of mold infections (below 5%) in combination with a mold-directed diagnostic approach based on high-resolution CT scan and biomarkers [94]. For Centers with a higher incidence of IFD, ECIL recommends either voriconazole (200 mg q12h) or itraconazole (BI recommendation), as they provide adequate protection; posaconazole receives a BII recommendation in this setting.

On the contrary for the USA based IDSA, posaconazole prophylaxis is recommended both during the pre-engraftment and

Table 1. Strength of recommendation on antifungal prophylaxis in allo-HSCTs.

	Pre-engraftment phase			Post-engraftment phase/GvHD		
	ECIL 2017 [94]	ESCMID 2018 [95]	IDSA 2017 [96]	ECIL 2017 [94]	ESCMID 2018 [95]	IDSA 2017 [96]
Fluconazole 400 mg QD	A I *	B II #	-	A III against	-	-
Itraconazole 400 mg QB	B I	D I	Strong R; mod-QE	B I	C II	Strong R; high-QE
Posaconazole oral sol 200 mg q8h	B II	B II	Strong R; high-QE	A I	A I	Strong R; high-QE
Tab 300 mg QD	B II	B II		A I	A I	
Voriconazole 200 mg q12h	B I	C I	Strong R; mod-QE	B I	C II	Strong R; mod-QE
Caspofungin 50 mg QD	No data	-	weak R; low-QE	-	-	C III
Micafungin 50 mg QD	B I LR-mold C I HR-mold	C I	weak R; low-QE	C II	C III	C III
L-AmB	C II	-	C III	C II	-	C III
Aerosol AmB	C III	-	-	-	-	C III

*centers with low incidence of mold infections, combined with a mold-directed diagnostic approach based on high-resolution CT scan and biomarker.

associated with liposomal amphotericin B.

Abbreviations: LR-mold, low ratio of mold infection; HR-mold, high ratio of mold infection; R, recommendation; QE, quality of evidence; Mod, moderate.

post-engraftment phases (oral solution 200 mg q8h or tablet formulation 300 mg q24h following a loading dose of 300 mg q12h on the first day) (Table 1), by extrapolating from the results of the study in neutropenic AML/MDS [94–96].

5. Viral infections

Viral infections after HSCT follow a specific timing with a prevalence of *Herpes simplex virus* (HSV) in the pre-engraftment phase, and *Cytomegalovirus* (CMV) together with *Adenovirus*, *Polyomavirus*, *Epstein-Barr virus* (EBV) and respiratory viruses in the post-engraftment phase. In the late phase *Varicella zoster virus* is the most frequently isolated virus [15]. Before transplantation, all patients should be tested for IgG antibodies against the most common virus that could reactivate during the period of immune deficiency, such as *Herpes simplex virus*, *Epstein-Barr virus*, *Cytomegalovirus*, *Varicella Zoster virus* and hepatitis.

Recommendations from the different international guidelines for viral infections prophylaxis in HSCT are summarized in Table 2.

5.1. Herpes simplex virus (HSV)

Herpes virus persists in the host after a primary infection and almost all recipients are seropositive before HSCT. Therefore, in the absence of antiviral prophylaxis, up to 80% of patients may experience a reactivation of HSV, during the early phase post- HSCT. Seropositive patients should receive acyclovir as antiviral prophylaxis to prevent HSV reactivation [11]. Usually, acyclovir has to be started together with the conditioning regimen and should be continued post-engraftment until the patients is receiving immunosuppressive therapy: the dose is usually 400 mg q8 h orally [97]. The duration of acyclovir prophylaxis may vary from center to center, but low dose acyclovir may be given for as long as 1-year post HSCT.

Table 2. Recommendation for prevention of viral infections.

Virus	Ref	Recommendations
Herpes Simplex virus	[97]	Prophylaxis with iv acyclovir 250 mg/mq or 5 mg/kg q12h (AI), oral acyclovir 200 q8h to 800 mg q12h (AI), oral valaciclovir 500 mg q12h p.o. (AII) or famciclovir 500 mg q12h p.o. (BIII) for 3–5 weeks after HSCT (BIII).
Varicella Zoster virus	[15,55,97]	Prophylaxis with oral acyclovir (800 mg q12h) or valaciclovir (500 mg q12-24 h) is recommended for seropositive patients, for 1 year (AII), or longer in case of GVHD and immunosuppression (BII). VZ-immunoglobuline (IG) should be administered within 96 hours after close contact with an infected person (AII). One dose of live-attenuated varicella vaccine is recommended in seronegative patients after a minimum of 24 months from HSCT in absence of GvHD, immunosuppression, relapse of the underlying disease and recently administration of immunoglobulin (BIIr).
Cytomegalovirus	[105,117,118,121–123]	Prophylaxis with high doses of aciclovir (CI) or valaciclovir (BI) reduced the risk of CMV infection. Letermovir significantly reduces CMV infection rate in CMV seropositive recipients (AI). Either intravenous ganciclovir or foscarnet can be used for first-line preemptive therapy (AI). Oral valganciclovir can be used in absence of severe gastrointestinal GvHD (AIIu).
Human Herpesvirus-6	[127]	Foscarnet (90 mg/Kg q12h) and ganciclovir (5 mg/kg q12h) were recommended for the treatment of HHV-6 related encephalitis post-HSCT, together with immunosuppression reduction, whenever possible (AIIu).
Epstein Barr virus	[132]	In patients with significant asymptomatic EBV-DNAemia, ECIL recommends the use of rituximab as preemptive approach (BIIu). A number from one to 4 weekly doses is considered appropriated until DNAemia negativity (AIIu), combined to reduction of immunosuppression if possible (AIIu). Prophylaxis with rituximab low dose may be considered in very high risk patients.
Adenovirus	[149]	A preemptive treatment with cidofovir (3–5 mg/Kg/week for 2–3 weeks and then every one weeks) is recommended by ECIL for high risk patients showing an increased viral load of DNA (BII).
Respiratory viruses	[11,173,173,183,183]	All viruses: Ivlg should be supplemented in patients with low serum IgG levels and recurrent infections associated with hypogammaglobulinemia (BIIr). Influenza: The ECIL group recommends one dose of seasonal inactivated influenza vaccine, yearly at the beginning of influenza season, and as long as the patient is assessed to be immunocompromised (AII). Influenza: First line therapy is oseltamivir at the dose of 75 mg bid for mild cases and 150 mg bid for severe cases for at least 10 days (BII). Current alternative treatment is inhaled zanamivir (B-II). RSV: Aerosolized ribavirin can be administered as 2 g every 8 hours or as 6 g per day for 7–10 days (BII). Also systemic ribavirin combined with immunoglobulin administration could be used (BIII). Parainfluenzae: control measures to reduce infections (AI). For patients with low respiratory tract involvement after HSCT treatment with aerosolized ribavirin combined with immunoglobulin infusion might be considered (BIII).
Hepatitis	[185,185,185,185,185]	HBV: Antiviral prophylaxis (lamivudine or entecavir) should be administered from the beginning of immunosuppression and maintained until 12 months after cessation of immunosuppression (BII). HBV: Vaccination (double vaccine dose) of anti-HBc negative and anti-HBs negative patients before HSCT and after HSCT (AI) is recommended; vaccination (accelerated single-dose three-weeks schedule) of anti-HBc negative and anti-HBs negative stem cell donors before stem cell harvest (BII) should be considered. HDV: Patients and stem cell donors anti-HBc positive should be also screened for HDV (AII). Patients with HBV and HDV co-infection should be managed as for HBV single infection (AI). HCV: Closely HVC-RNA load monitoring is recommended (AIII). Expert liver monitoring is recommended for HCV infected recipient after HSCT (AIII). Antiviral treatment should be considered for HCV-RNA positive subject once the hematological disease has been brought under control (AIII). HAV: avoid HSCT in viremic recipient or donors (AIII) and to administer vaccination to HAV seronegative recipient of HSCT (BII). HEV: For patients with chronic HEV infection, a reduction in immunosuppressive drug should be considered when possible (BIII). Oral ribavirin could be safely and efficaciously administered in HEV-RNA positive patients who had obtained a stable engraftment after HSCT (BIII).

When acyclovir is administered for CMV prophylaxis, the dose is significantly higher, as will be detailed [97].

5.2. Varicella zoster virus (VZV)

After primary infection from *Varicella Zoster Virus* (VZV), the virus establishes latency in the dorsal ganglia and may be reactivated over time as *Herpes zoster* in immunocompetent individuals or as severe varicella episode in immunocompromised patients. Corticosteroids administration as well as immunosuppressive therapy, chronic GvHD, cord blood graft, a diagnosis of lymphoproliferative disease, and alternative donor transplants, all represent risk factors for VZV reactivation after HSCT [98–102]. Prophylaxis with acyclovir (800 mg po q12h) or valaciclovir (500 mg q24–12 h) is recommended for seropositive HSCT recipients, for 1 year, or longer in case of GvHD or immunosuppressive therapy [97]. Immunocompromised HSCT recipients should receive VZV-immunoglobulin (VZIG) within 96 hours after close contact with an infected person [97]. Postexposure acyclovir or valacyclovir may be used as an alternative [15]. VZIG, acyclovir, or valacyclovir should also be administered to all VZV-seronegative HSCT recipients undergoing conditioning therapy who are exposed to a VZV vaccine with rash. If VZV-seronegative HSCT recipients are closely exposed to varicella more than 3 weeks after VZIG administration, they should receive another dose of VZIG [103]. Vaccination is regarded in the ECIL report as an option: a single dose of live-attenuated varicella vaccine in seronegative patients after a minimum of 24 months from HSCT in absence of GvHD, immunosuppression, relapse of the underlying disease and recently administration of immunoglobulin in the past 8 months [55]. A recent report with recombinant VZV vaccine appears to be very encouraging [104].

5.3. Cytomegalovirus (CMV)

Cytomegalovirus (CMV) is one of the most frequent viral infections detectable after HSCT. Historically, CMV disease developed in 10–40% of patients, frequently as a pneumonia, with a mortality rate of 70%, or as a gastrointestinal disease [105–107]. Things have changed with the advent of early diagnosis of CMV viremia, first with antigenemia and then with a PCR based approach, and preemptive treatment with either ganciclovir or foscarnet [105,106]. As a result there has been a drastic reduction of CMV disease, and of CMV associated mortality [105,106]. The risk factors for CMV infection after HSCT are as follows: positive serological status of both the donor and the recipient, high-dose corticosteroids, T-cell depletion, GVHD, and the use of a donor other than an HLA identical sibling [108–110]. The CMV DNA load can be high, and response to treatment not satisfactory, with long lasting viremia, in CMV positive patients, receiving a graft from a CMV negative donors [105]. The recipients' seropositivity is an adverse factor for non-relapse mortality [111,112].

The ECIL-7 guidelines recommend testing patients and donor for CMV IgG antibodies close to the time of HSCT,

using highly sensitive assays. In addition, a CMV seronegative donor should be selected for a seronegative recipient, and a CMV positive donor, for a CMV positive patient [105].

To prevent primary CMV infection for CMV seronegative recipient, blood products from seronegative donors or leucocyte-depleted blood products should be used [113–116]. Antiviral CMV prophylaxis has been tested in seropositive recipients. High doses of acyclovir or valaciclovir reduce the risk of CMV infection [117,118], and high dose acyclovir also improves survival [117]. Ganciclovir prophylaxis reduces the risk of CMV disease but is associated with significant toxicity [119,120]. Recently, letermovir, has been shown to reduce clinically relevant CMV infections (37%) in CMV seropositive recipients, as compared to a placebo group (61%), without relevant side effects [121–123]. Prophylaxis with letermovir appeared superior to prophylaxis with other agents in reducing all cause mortality [122]. So should we give letermovir to all patients undergoing an allogeneic HSCT? A consensus conference was convened to discuss this issue, and the recommendation is to use letermovir prophylaxis, until day +100 post HSCT, in CMV IgG positive recipients [123]. High dose acyclovir should be given to other recipients.

5.4. Herpes virus-6 (HHV6)

Herpesvirus 6 (HHV-6) reactivation is common during the early transplant period, with viremia occurring in approximately 40–60% of patients [124–126]. There are two different species of HHV-6, A and B. No disease has been linked to HHV-6A, whereas HHV-6B is the most frequent cause of encephalitis post-HSCT. Major risk factors for HHV-6B encephalitis are the presence of HHV-6B viremia and the use of cord blood as a stem cell source [127–130]. Screening for HHV-6B DNAemia and prompt identification of clinical symptoms, rather than routine screening, is crucial for the diagnosis [131]. No prophylaxis is suggested. Foscarnet (90 mg/Kg bid) and ganciclovir (5 mg/kg bid) are recommended for the treatment of HHV-6 related encephalitis post-HSCT, together with reduction of immunosuppressive therapy, whenever possible [127].

5.5. Epstein Barr virus

The incidence of EBV-DNAemia varies between transplant centers ranging from 0.1 to 63% [132]. A recent study has defined a risk score for developing EBV lymphoproliferative disorders [133], with the following risk factors: a diagnosis of aplastic anemia, high dose ATG in the conditioning regimen, cord blood or mismatched unrelated donor grafts. For patients at high risk primary prophylaxis with a low dose of rituximab is effective in reducing the risk of EBV reactivation [134]; this is particularly true for patients with aplastic anemia undergoing a cord blood transplant [135]. Low dose rituximab (200 mg) prophylaxis, given on day +5 after transplant, reduces EBV reactivation in high-risk patients [134,135]. For standard risk patients, monitoring EBV DNA in whole blood [136], should be performed weekly, and preemptive rituximab treatment should be considered for patients with a high EBV DNA copy

number [137–139]. The use of cytotoxic T cells (EBV-CTLs) for prevention and treatment of PTLD is effective but requires a dedicated laboratory and expertise [140]. The incidence of PTLD varies from 1.2% in matched related donor to 11.2% in mismatch unrelated donors [132]. Prolonged immunosuppression, cord blood graft, previous splenectomy, second HSCT, HLA mismatch and EBV seropositive status of the donor are reported as pre-transplant risk factors for PTLD [141–145]. Other post-transplant risk factors, are prolonged and intensive immunosuppression for GVHD treatment and a high EBV viral load [97,144–148]. Prophylaxis and treatment of EBV-PTLD includes rituximab use, reduction of immunosuppression, EBV-CTLs and chemotherapy. Rituximab controls approximately 90% of patients with EBV-DNAemia and 65% of patients with overt PTLD. Reduction of immunosuppression leads to a favorable outcome in 68% of the cases when used alone and in 80% when combined to rituximab administration. Finally, the use of EBV-CTLs improves the outcome in 75% of EBV-DNAemia and over 90% of PTLD. ECIL has no recommendation for EBV prophylaxis, however, for patients with significant EBV-DNAemia, without clinical symptoms the use of rituximab is recommended as a preemptive approach [132].

5.6. Adenovirus

Adenovirus viremia after HSCT might remain asymptomatic in 50–80% of the cases [148–150], whereas 30 to 60% of patients with stool positivity have enteric symptoms [151,152]. The clinical manifestation of adenovirus ranged from mild gastroenteritis to pneumonia, hemorrhagic enteritis and urinary manifestations, hepatitis and organ failure [148]. High-risk patients are those receiving T-cell depleted grafts or HLA mismatched donor transplants, patients with GVHD on corticosteroids, and those receiving cord blood stem cells [151,153–157]. Patients at high risk for adenovirus infection should be monitored with quantitative PCR test during the first 6 months [158]. Anti-viral prophylaxis is not recommended. The clearance of adenovirus was seen to be associated with immune reconstitution, with a longer viral shedding in lymphopenic patients [158–160].

5.7. Respiratory virus

In the last years an exponential increase in respiratory viral infections has been documented in allogeneic HSCT recipients. The main viruses involved are respiratory syncytial virus (RSV), influenza, parainfluenza, metapneumovirus, and adenovirus [11,15,160]. The recommendation is to avoid contacts with individuals showing signs and symptoms of respiratory infections [161–163]. Although prophylactic immune globulin infusions are not recommended as a routine practice for HSCT patients [164,165]; nevertheless, immune globulins should be infused in patients with low serum IgG levels and recurrent infections [11].

Influenza is a common infection, and approximately 30% of positive patients develop serious lower respiratory tract symptoms, with a high mortality rate [166,167]. The main risk factors for low respiratory tract influenza infection, after HSCT,

are the following: lymphopenia, myeloablative conditioning as compared to non-myeloablative one, chronic steroid treatment and older age [166,168–170]. Lymphopenia and corticosteroid administration might cause a prolonged viral shedding after infection [167,171,172]. Qualitative reverse transcriptase PCR (RT-PCR) on nasal or throat swab or broncho-alveolar lavage (BAL) is the most sensitive and specific test to identify respiratory viruses infection [172]. The most effective prophylaxis for HSCT patients is vaccination. The ECIL group recommends one dose of seasonal inactivated influenza vaccine, yearly at the beginning of the influenza season, and as long as the patient is assessed to be immunocompromised. A second dose after 3–4 weeks might be beneficial for high risk patients. They also recommended to extend flu vaccination to family members and household contacts [55]. Vaccination response rate was of 10 to 40% after one dose of inactivated influenza vaccine administered 6 months after HSCT, but the efficacy might be different according to the seasonal strain, and to patients' characteristics such as lymphopenia, hypogammaglobulinemia, GVHD, immunosuppressive treatment and rituximab administration [173–178].

Other community acquired respiratory virus (CARVs), other than adenovirus and influenza, are diagnosed by RT-PCR performed on nasal swab or BAL [179]. Respiratory syncytial virus occurs in 1–2% of patients, and causes low respiratory tract involvement in one third of cases [180–182]. Risk factors are the same reported for influenza virus. Aerosolized ribavirin can be administered as 2 g every 8 hours or as 6 g per day for 7–10 days; systemic ribavirin combined with immunoglobulin administration could be used [183].

Human parainfluenza virus might cause symptomatic infection in 2 to 7% of cases, a third of which with involvement of the low respiratory tract. After HSCT, approximately 3–9% of patients experienced human *Metapneumovirus* infection, 5.7% *Coronaviruses* infection, and 22% rhinovirus [184]. Best prevention against CARVs infection is appropriate behavioral measures: good personal hygiene, hand washing, covering mouth and nose when coughing and sneezing, avoiding contacts with individuals with symptoms of a respiratory tract infection, isolation room and application of strict protection measures for HCWs and visitors [183].

5.8. Hepatitis

Patients eligible for HSCT should be screened for hepatotropic viruses before chemotherapy. In the same way, all donor should be evaluated for hepatotropic viruses before stem cell harvest given that an infected donor can transmit viral hepatitis [185].

For B hepatitis the risk of severe hepatic complication is very high in HBsAg positive patients, as well as for HBsAg negative and anti-HBc positive subjects [185]. Other risk factors are a low titer of anti-HBs antibodies, the use of conditioning regimen combining different drugs, GVHD and prolonged immunosuppression [186,187]. Nucleoside analogues, lamivudine and entecavir are currently used for HBV

treatment in patients undergoing HSCT, found positive for HBV [188]. The ECIL guidelines recommends to start antiviral drugs prophylaxis when immunosuppression is initiated, and to maintain patients under treatment for at least 12 months after discontinuation of immunosuppressive therapy [185]. However, late reactivations after HSCT are reported, especially in patients who had received anti-CD20 monoclonal antibody [189]. Antivirals should be given for longer periods also in HSCT recipients with chronic GvHD [185].

Vaccination (double vaccine dose) of anti-HBc negative and anti-HBs negative patients before HSCT and after HSCT is recommended; vaccination (accelerated single-dose three-weeks schedule) of anti-HBc negative and anti-HBs negative stem cell donors before stem cell harvest should be considered [185]. Patients and stem cell donors anti-HBc positive should also be screened for HDV. HSCT is not recommended if donor harbors HBsAg and anti-HDV antibodies or HDV RNA. Patients with HBV and HDV co-infection should be managed as for HBV single infection [185].

Patients with HCV infection in the setting of HSCT seem to have a worse outcome, especially those with liver fibrosis, who are at high risk for sinusoidal obstruction syndrome (SOS) or veno-occlusive disease (VOD). Close monitoring of the HCV-RNA load is recommended, and increasing HCV RNA load must be promptly recognized [185].

Patients receiving HSCT might be at risk of acquiring HEV through blood transfusion or the stem cell graft [190]. The lack of anti-HEV specific T cells combined with calcineurin inhibitors administration, may favor HEV replication [191]. For patients with chronic HEV infection, a reduction of immunosuppressive therapy should be considered, when possible [185].

Even if blood products are tested for HAV, there is a risk of HAV infection for HSCT patients [192]. In these patients HAV might cause aplastic anemia and SOS [193,194]. The ECIL recommendations is to avoid HSCT in viremic recipient or donors, and to vaccinate HAV seronegative recipients [185].

5.9. New emergency due to SARS-CoV2

The European Group for Blood and Marrow Transplantation (EBMT) has issued interim guidelines for recipients and donors of HSCT in the SARS-CoV2 pandemic era [195]. The recommendations can be summarized as follows:

- all recipients should test negative for SARS-CoV2, before starting the conditioning regimen
- all recipients are advised to have a 14 day home isolation before admission
- if a recipient proves SARS-CoV2 positive, the transplant should be deferred

The World Marrow Donor Association (WMDA) has issued guidelines for donors [196]:

- SARS-COV2 positive donors must be excluded from donation

- if a donor gets in contact with a SARS-CoV2 positive individual, the donation should be deferred for at least 28 days. If the transplant is urgent and the donor asymptomatic, an earlier collection may be considered.
- due to air traffic problems, the stem cell harvest should preferably be sent to the transplant Centers, before starting the conditioning regimen, and cryopreserved.

6. Expert opinion

Infectious complications after allogeneic HSCT come in 3 distinct periods: an early neutropenic phase (day +1 + 30), an intermediate period of early immuno-reconstitution (day +30 + 100) and a late period (beyond day +100).

The first period is clearly dominated by bacterial infections, with significant morbidity and mortality. Bacterial infections remain an open issue, especially because of the emergence of antibiotic-resistant strains: prevention or reduction of infections due to MDR bacteria relies on global hospital strategies, compliance with behavioral procedures and on the allocation of financial resources. In the absence of any of these, the outcome will be negative, and further dissemination of MDR bacteria will occur. Over the years we have reduced the toxicity of the conditioning regimen, and have thus reduced tissue damage: however the advent of high dose cyclophosphamide post-transplant (PT-CY), an extremely effective way to prevent GvHD, has unfortunately increased gut toxicity, and associated early BSI. Transplantation is a package and changes in one of the package components leads to changes in outcome. Because the gut has such a predominant role, the microbiota has received considerable attention: specific microbiota signatures can be used to identify patients at risk, eligible for tailored prophylaxis. In addition diversity in the microbiome has recently been shown to be a strong predictor of mortality in allogeneic transplant recipients [197].

Fungal and viral infections dominate the second post-transplant period. GvHD is a major determinant of immune-reconstitution: therefore effective prevention of GvHD will result in less GvHD and improved hematologic and immunologic reconstitution. For unmanipulated stem cell transplants, we are currently moving from standard prophylaxis (a calcineurin inhibitor and methotrexate with anti-thymocyte globulin) to post-transplant cyclophosphamide: the latter has been used in the past decade only for mismatched transplants, but we are now starting to test its efficacy in matched transplant. We have already seen a reduction of GvHD and improved immune reconstitution, and we expect to see further improved outcome. Letermovir is the other new player in this period: effective prevention of CMV, with very low toxicity appears to be a significant bonus, which may further contribute to low alloreactivity, low inflammatory reactions and low rate of infectious complications. EBV we have learned to monitor and treat preemptively, and we can now identify patients at very high risk, such as patients with aplastic anemia

undergoing a cord blood transplant with anti-thymocyte globulin in the conditioning regimen: for these patients prophylactic rituximab is recommended. For respiratory virus, to date, the only way to reduce potentially life-threatening complication, is avoiding contacts with symptomatic subjects, social distance we would say today. In this sense it might be helpful to intensify the vaccination campaign among healthy people, in addition to HSCT recipient and relatives, given the spread of respiratory viruses in community.

As to antifungal agents, issues remain on drug-drug interactions, daily administration, toxicity and variable exposure of azoles, echinocandins and polyenes. Rezafungin is a new broad-spectrum echinocandin (with demonstrated potent *in vivo* activity against some representative strains of *Candida* spp, *Aspergillus* spp, *Pneumocystis* spp and dermatophytes): it is under development. Rezafungin is derived from anidulafungin, with a long half-life (80 h) and a favorable safety profile that allows for high doses given once weekly. A phase 3 trial for treatment of candidemia and invasive candidiasis is ongoing, as well as another phase 3 for *Candida*, *Pneumocystis jirovecii* and *Aspergillus* in HSCT. Itraconazole and voriconazole inhaled solutions may produce high drug concentrations in the lung, while avoiding toxicities associated with systemically administered agents. Pharmacokinetics and efficacy of nebulized voriconazole have been described in murine models some years ago and some recent *in vivo* studies about biodistribution and deposition of these drug-nanoparticles, suggested they could be a therapeutic choice, even when compared to the intravenous administration of the free drug.

In the late period again GvHD, in its chronic form, together with its treatment, plays a crucial role in determining whether a patient will or will not develop infectious complications. Patients with moderate/severe cGvHD are exposed to bacterial, fungal and viral infections, and will require constant monitoring and frequent hospital admission. Moderate severe cGvHD is an untreatable disease, with success rates (clearing GvHD) in the order of 15%. Patients without cGvHD, can discontinue immunosuppressive therapy (calcineurin inhibitor) by day +180 and are usually free of infectious complications, leading a normal life. Therefore the key imperative message is: prevent chronic GvHD. Changes in prophylaxis for matched transplants, with high-dose cyclophosphamide, are expected to lead to a major reduction in acute and chronic GvHD, and related complications.

Declaration of interest

The authors have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

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